

Question framing and context source modulate scalar implicature rates in verification tasks

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Scalar implicatures (SIs) are commonly studied using sentence verification tasks. In these tasks, participants judge underinformative sentences such as *Some trees are plants*. A standard linking assumption holds that rejecting such sentences as false reflects a pragmatic interpretation (but see [1]). Accepting them as true reflects a literal interpretation [2, 3].

Literal vs. pragmatic response rates for the scalar *some* vary widely across studies (e.g., from 38% literal responses [4] up to 77% literal responses [5]). Prior work has shown that differences in task demands play a crucial role in shaping observed response rates in both adults and children [6, 7, 8, 9, 10].

A still somewhat underexplored source of task-related variation is question framing, i.e., whether judgments are about truth (*true/false*) or about felicity (*good/bad*), as remarked by [1]. Another is the source of context against which to verify, i.e., either one's own world knowledge (WK) or a provided picture (sentence–picture verification, [11]). We tested the impact of these two factors on SI response behavior.

Across four web-based experiments (N = 252, 252, 576, and 576), we used a 2×2 between-subjects design. We manipulated question framing (truth vs. felicity) and context source (WK vs. picture). Participants judged underinformative target sentences as well as control sentences, responding with either *true/false* or *good/bad* judgments. **Figure 1** illustrates an example item. There were 24 items overall.

Experiments 1 and 2 included 24 trials per participant, evenly distributed across three conditions: target, control true, and control false. Experiment 3 also included 24 trials per participant but featured three additional control types using the stronger scalemate *all*. Further, Experiment 2 differed from the other two in response format: In Experiments 1 and 3, participants chose between explicit polarity pairs (e.g., *Is this a good or a bad statement?*). In Experiment 2, they answered *yes/no* questions instead (e.g., *Is this a good statement?*). Experiment 4 was a direct replication of Experiment 3, but using French rather than English (with the French scalar *certain/certaines*).

All four experiments showed the same basic result pattern. Participants in the truth condition gave more literal responses than those who were in the felicity condition (**Exp. 1:** $b = 4.3$, $SE = 1.0$, $z = 4.4$, $p < .001$; **Exp. 2:** $b = 2.9$, $SE = 0.8$, $z = 3.7$, $p < .001$; **Exp. 3:** $b = 3.0$, $SE = 0.4$, $z = 7.1$, $p < .001$; **Exp. 4:** $b = 1.0$, $SE = 0.4$, $z = 2.7$, $p = .006$). We also observed a consistent main effect of context source: Participants in the WK-based condition responded more literally than those in the picture-based condition (**Exp. 1:** $b = 4.5$, $SE = 1.0$, $z = 4.5$, $p < .001$; **Exp. 2:** $b = 1.8$, $SE = 0.8$, $z = 2.3$, $p = .020$; **Exp. 3:** $b = 2.9$, $SE = 0.4$, $z = 6.8$, $p < .001$; **Exp. 4:** $b = 0.8$, $SE = 0.4$, $z = 2.1$, $p = .036$). Note, however, that effect magnitudes were generally smaller in French (Exp. 4) than in English (Exp. 1–3). **Figure 2** displays the average literal response rates across conditions and experiments. These results suggest that SI processing is indeed substantially modulated by question framing and context source.

More broadly, our findings support the idea that seemingly superficial choices in task design can systematically influence how underinformative sentences are interpreted. Future work should identify which design factors most reliably affect literal vs. pragmatic

response rates. A better understanding of how contextual features shape response rates will help clarify which aspects of SI processing are stable and which depend on particularities of the experimental setup in which they are assessed.

Experiment scripts for PClbex [12], the collected response data, and R code for statistical analysis can be accessed here (anonymized link):

- https://osf.io/86r4u/files/osfstorage?view_only=61773f9ca6e84e5db09e317d57617947

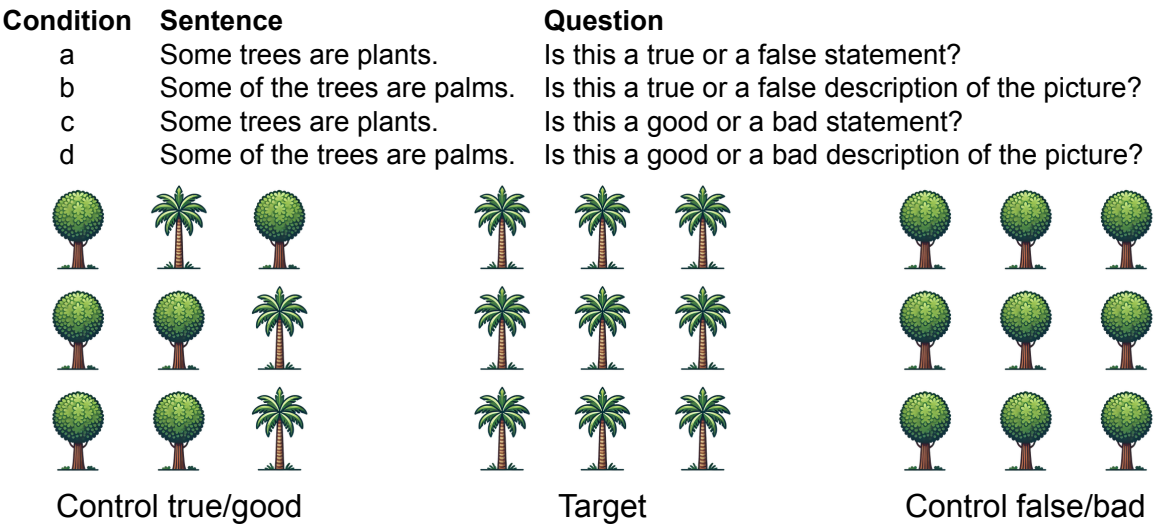


Figure 1: Above: Example item shown in each of the four conditions in the 2×2 design. Conditions a and b use world knowledge as context; c and d use pictures. Conditions a and c use truth-based framing; b and d use felicity-based framing. Below: Pictures for the example item. For the sentence *Some of the trees are palms*, the control picture on the left requires an unambiguous *true/good* verification. The target picture in the middle is pragmatically ambiguous. The control picture on the right requires an unambiguous *false/bad* verification.

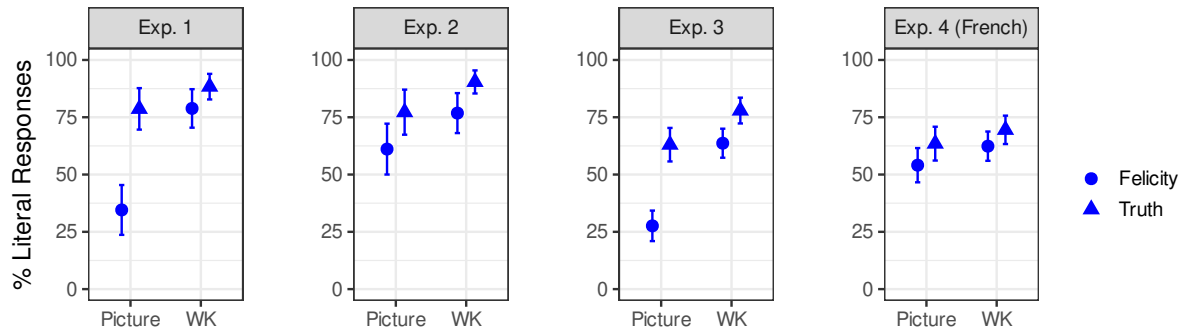


Figure 2: Average literal response rates across the four conditions in Experiments 1 to 4. Error bars represent 95% confidence intervals computed over subject-level means.

References

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